VIVEKANANDA GLOBAL UNIVERSITY, JAIPUR

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Assignment III

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Unit III

Subject: Database Management System

Subject Code : UGCSE211

Department of Computer Science & Engineering

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Assignment: Module-iii Relational Model & SQL

3	**Module 3: Relational Model & SQL** **(9L)** **Relational Model:** Structure of relational Databases, Relational Algebra, Relational Calculus, Extended Relational Algebra. **SQL:** Form of Basic SQL Query, Nested Queries, Aggregative Operators, NULL values, Logical operators, Outer Joins, Complex Integrity Constraints in SQL.

Part A: Q1. Relational Algebra

1. Explain the following operations with examples:
   - Selection (σ)
   - Projection (π)
   - Union (∪)
   - Set Difference (−)
2. What is **Join operation**? Explain types.

Q2. Relational Calculus

1. What is **Relational Calculus**?
2. Difference between:
   - Tuple Relational Calculus (TRC)
   - Domain Relational Calculus (DRC)

Part B: SQL Practical (Table-Based Questions)

Table 1: Student

StudentID	Name	Age	Marks	City
1	Ravi	20	85	Delhi
2	Aman	21	78	Mumbai
3	Neha	19	92	Delhi
4	Sita	22	88	Pune
5	Raj	20	NULL	Mumbai

Questions on Student Table

1. Write a query to display all students.
2. Show names of students who belong to **Delhi**.
3. Find students with marks **greater than 80**.
4. Display students whose marks are **NULL**.
5. Find **average marks**.
6. Find **maximum and minimum marks**.
7. Count total number of students.
8. Display students in **descending order of marks**.
9. Show only **Name and Marks** columns.
10. Find students whose name starts with **'R'**.

Table 2: Employee

EmpID	Name	Salary	DeptID
101	Rahul	50000	1
102	Priya	60000	2
103	Amit	55000	1
104	Sara	70000	3

📊 **Table 3: Department**

DeptID	DeptName
1	HR
2	IT
3	Finance

Join Queries

1. Write query to display employee name with department name.
2. Use **INNER JOIN** to show matching records.
3. Use **LEFT JOIN** to show all employees with department.
4. Find employees working in **IT department**.

Nested Queries

1. Find employees earning more than **average salary**.
2. Find employee with **highest salary**.
3. Display employees whose salary is greater than **Rahul's salary**.

Aggregate Queries

1. Find total salary of all employees.
2. Find average salary department-wise.
3. Count employees in each department.

Logical Operator Queries

1. Find employees with salary between **50000 and 70000**.
2. Find employees in department 1 **OR** salary > 60000.
3. Find employees **NOT** in department 2.

Outer Join Queries

1. Show all departments even if no employee exists.
2. Show all employees even if department is missing.